

Dissecting Corporate Culture Using Generative AI

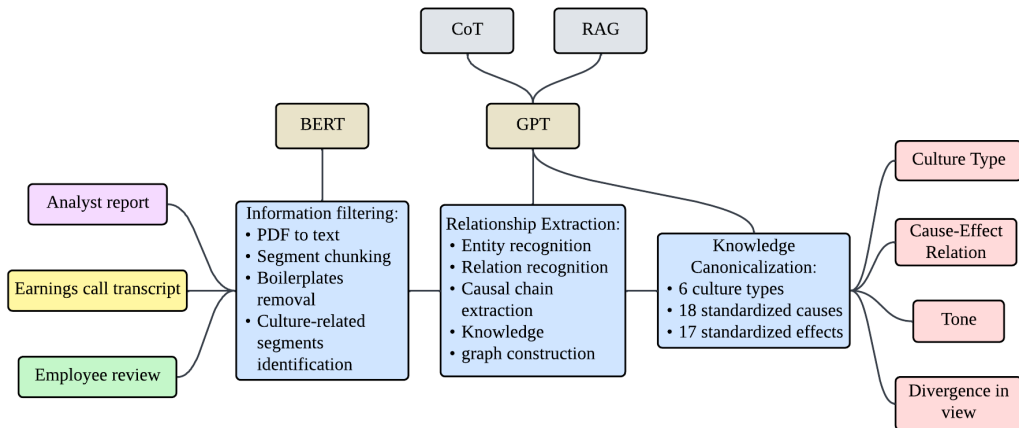
Kai Li, Feng Mai, Rui Shen, Chelsea Yang, Tengfei Zhang (2026, *RFS*)

解读人：李子明

武汉大学金融系

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Overview



Motivation

- Corporate culture in perspectives among different stakeholders
 - Growing recognition of culture as important determinant of corporate behavior stimulate cultural revolution in finance (Zingales, 2015).
 - Existing studies largely treat corporate culture as homogeneous organizational attribute.
 - Divergence in perceptions and attitudes across different stakeholder groups unexplored.
- Generative AI for corporate culture analysis
 - Different stakeholders express through heterogeneous communication venues (corporate website, executive survey, analyst report, earnings call, employee review, job posting).
 - Prior studies predominantly rely on single information source (Grennan and Li, 2023).
 - Integrating and aligning multiple large-scale textual datasets remain challenging.

Research Question

- Which cultural values are important to different stakeholders in a modern corporation?
- How does corporate culture matter according to different stakeholders?
- What explains the divergence in perspectives among different stakeholders?
- What is the relationship between corporate culture and price formation?

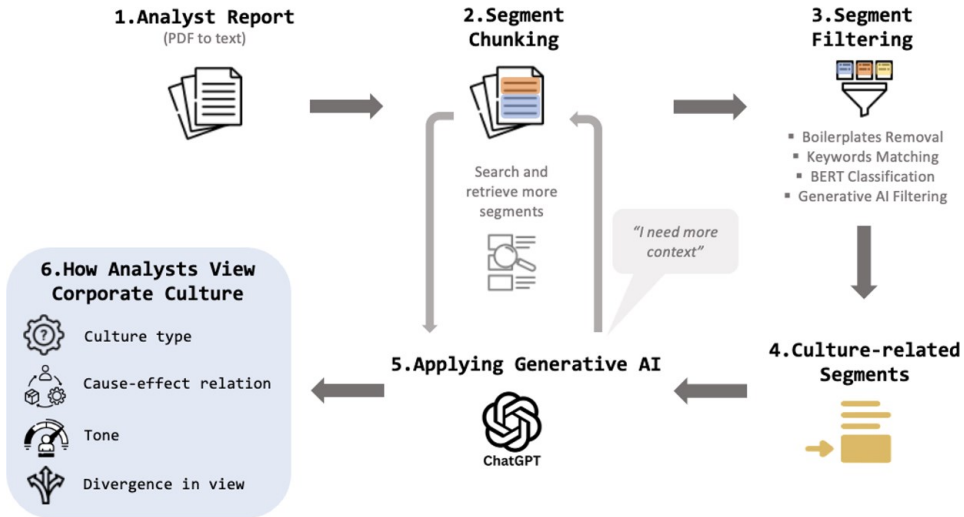
Contribution

- Contribute to literature on corporate culture.
 - Prior: single group's perspective on culture (Guiso et al., 2019; Grennan, 2019; Au et al., 2021; Briscoe-Tran, 2022; Graham et al., 2022; Li et al., 2024).
 - This paper: three distinct stakeholder groups' perspectives using uniform approach.
- Contribute to literature on big data and machine learning in finance.
 - Prior: generative AI derive new information and glean valuable insights (Bai et al., 2023; Bybee, 2023; Jha et al., 2023; Kim, 2023; Li, 2023; Lopez-Lira, 2023).
 - This paper: combine versatility of LLM with efficiency of SLM; apply for structured information extraction rather than text classification or prediction task.
- Contribute to literature on equity analysts.
 - Prior: textual analysis to analyst reports (Asquith et al., 2005; Bellstam et al. 2021), unable to unpack "black box" of analysts' fundamental research.
 - This paper: elucidate process of transforming soft information into actionable insights.

Textual Data Sets

- Analyst report
 - 2.4 million analyst reports from Thomson One's Investext database (2000 - 2020)
- Earnings call transcript
 - 243 thousand earnings calls from Capital IQ Transcripts database (2004 - 2020)
- Employee review
 - 5.3 million Glassdoor employee reviews from Revelio Labs (2008 - 2020)

Flowchart of Information Extraction Method



Information Filtering

- PDF to Text
 - GROBID to transform PDF into XML and strip tables, annexes, notes.
 - OpenNLP to split plain text into sentences.
- Segment Chunking
 - Loss of structure and inability to distinguish between headers, bullet points, paragraphs
 - C99 to coalesce sentences into segments that align closely with coherent thoughts in text.
- Removing Boilerplate Segments
 - Select positive and negative example segments as training set
 - Deploy MLP head (Freeze parameters of BERT) to classify segments as boilerplate
- Identifying Culture-related Segments
 - Keyword search using terms explicitly related to corporate culture.
 - Fine-tune BERT to identify culture-related segments lack specific keywords.
 - Prompt instruct GPT to filter and retain segments substantively discuss culture.

Cause-Effect Relationship Extraction

- Reasoning agents
 - Main generative model: GPT-4o mini
 - CoT: guide GPT to break task into steps and systematically analyze different aspects.
 - RAG: allow GPT to request additional context if needed and retrieve relevant segments.
- Extraction
 - 6 culture types predefined by pilot study that capture main themes in discussion
 - Instruction: (1)summarize and classify culture; (2)determine tone; (3)analyze causality; (4)identify cause and effect; (5)extract causal graph triple [*Entity*₁, *Relation*, *Entity*₂]
- Canonicalization
 - Two-stage process to map phrases representing causes and effects of each culture to standardized entity (18 standardized causes and 17 standardized effects).
 - Instruction: (1)map to appropriate entity; (2)canonicalize cause-effect relation triples

Chain-of-Thought Prompt

As an expert specializing in corporate culture and causal reasoning, your task is to analyze segments from analyst reports about corporate culture. Your goal is to extract and interpret information about a company's corporate culture and identify cause-effect relationships. Present your findings in a structured JSON format. Let's think step by step.

Step-by-Step Instructions:

1. Summarize the Corporate Culture (Q1):

- Task: Determine the specific corporate culture being discussed in the segment.
- Action: Summarize it in a short phrase starting with an adjective. If corporate culture is not explicitly mentioned, infer it from the context.
- Note: Avoid using generic adjectives such as strong/weak or positive/negative culture. If more context from the report is needed for the analysis, output "I need more context" in the relevant JSON field.

2. Classify the Corporate Culture (Q2):

- Task: Categorize the identified corporate culture into one of the following six types:
 - * Collaboration and People-Focused: Focusing on (or deficient in) collaboration, cooperation, teamwork, supportive, low levels of conflict, community, communication within an organization, employee well-being, employee equity sharing and compensation, diversity, inclusion, empowerment, or talent.
 - * Customer-Oriented: Focusing on (or deficient in) sales, customer, customer service, listening to the customer, customer retention, customer experience, customer satisfaction, user experience, client service, being brand-driven, quality of product, quality of service, quality of solution, or taking pride in service.
 - * Innovation and Adaptability: Focusing on (or deficient in) innovation, creativity, technology, entrepreneurship, adaptability, transformations, flexibility, agility, willingness to experiment, beyond tradition, disruption, fast-moving, quick to take advantage of opportunities, resilience to change, or taking initiative.

Prompt to Canonicalize Cause-Effect Relation

As an expert specializing in corporate culture and causal reasoning, conduct entity canonicalization on selected phrases taken from analyst reports. These phrases are considered to be *causes (consequences) of corporate culture*. Your objective is to map each phrase to the most appropriate standardized entity name from the predefined list below.

Instructions:

- Review the provided examples for each standardized entity to grasp their scope and nuances.
 - Carefully analyze each phrase and assign it to the most fitting standardized entity based on its underlying meaning.
 - Assign the phrase to "Other" only if it does not align with any of the predefined entities.
- Before assigning a phrase to "Other", ensure that it does not reasonably fit any existing categories.

The available standardized entity names and examples are:

[Categories and examples of causes (see Table IA4 for details)]

JSON Output Structure:

```
{
  "all_results": [
    {
      "input_phrase_id": 1,
      "input_phrase": "",
      "canonical_entity": "One of the predefined standardized entity names or 'Other'"
    },
  ],
}
```

Examples of Extracted Cause or Effect Relations

Example 1: Coca-Cola's CEO Neville Isdell presented for the company at the CAGNY conference in Arizona this morning. While there was not much new news in the presentation, the company's tone has changed meaningfully from Isdell's presentation at CAGNY two years ago (when KO was promising the market very little) to today, when the company has greater confidence that its long-term algorithm is both working and sustainable. We highlight what we think were a few of the key takeaways below:

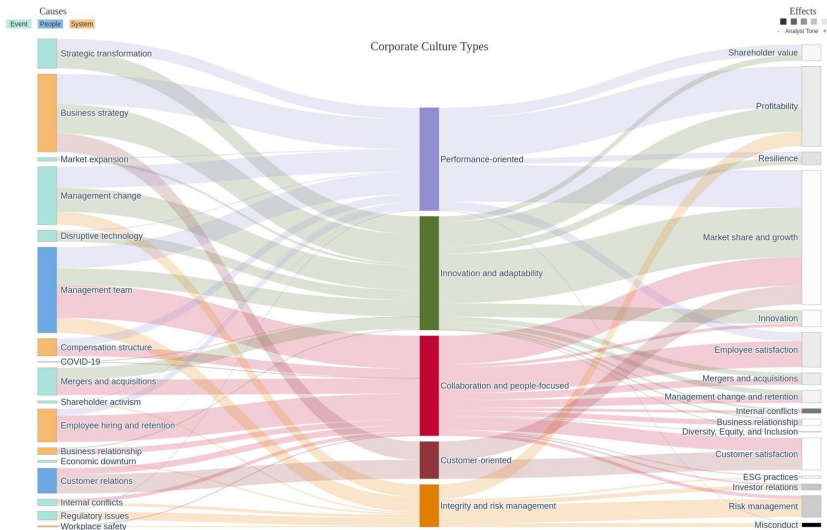
- The company highlighted the improving performance in 2006 as KO has moved through its Manifesto For Growth strategy and now enters a phase of likely sustainable growth with a focus on growing the core brands, capturing emerging platforms by establishing a culture of innovation, and providing franchise leadership to the bottlers (which was consistent with what we heard from CCE yesterday, with a greater focus on increasing collaboration with the bottling system).
- KO is entering 2007 with some of the strongest growth momentum in the last several years, and the company has demonstrated that they can continue to post solid growth despite underperformance in key markets.NA should remain a challenge in 2007, but KO has clearly shown an ability to turn around problem markets and seems very comfortable with its long-term growth model given strength in the balance of the business. While it's not yet clear what stage in the turnaround we have entered for key markets such as Japan, Germany, and India, KO expressed confidence in the recent improvements in these areas and we are encouraged by the results...

Extracted relation(s):

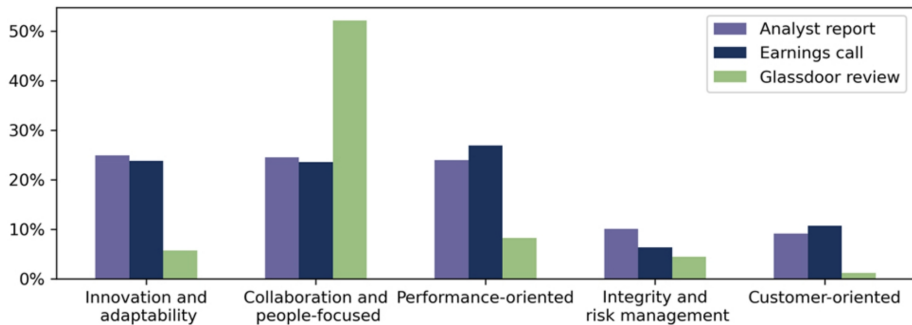
- **Manifesto for growth strategy (Business strategy) → Innovative and growth-oriented culture (Innovation).** Explanation: The culture is shaped by the strategic focus on growth and innovation.
- **Innovative and growth-oriented culture (Innovation) → Sustainable growth (Market share and growth).** Explanation: The culture fosters sustainable growth as indicated by the company's performance.

Ref: This report was written by John A. Faucher from JP Morgan for Coca-Cola Co. released on 2/22/2007.

Cause-Effect Knowledge Graph of Corporate Culture



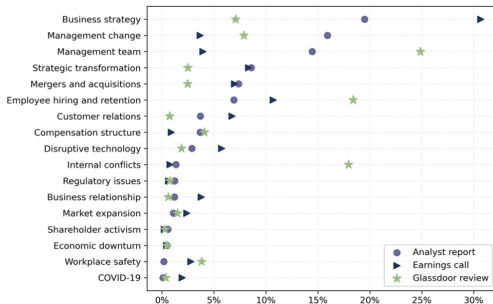
Different Stakeholder Groups' Perspectives on Culture Types



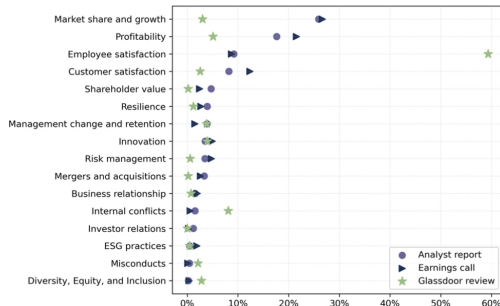
- Culture valued by different group consistent with their lived experiences
 - Analyst & Management: innovation and adaptability, performance-oriented, customer-oriented
 - Employee: collaboration and people-focused

Different Stakeholder Groups' Perspectives on Cause and Effect

B Different stakeholder groups' perspectives on causes of cultural change



C Different stakeholder groups' perspectives on effects of culture



- Analyst's narratives largely balance perspective of management and employee.
 - Frequency appear in analyst report fall between those in earnings call and employee review for majority of cause and effect.

Explaining Divergent Perspectives Held by Different Groups

- Euclidean distance to measure divergence
 - Analyst's frequency vectors for discussing five culture types: $\mathbf{A} = (a_1, a_2, a_3, a_4, a_5)$
 - Management's frequency vectors: $\mathbf{M} = (m_1, m_2, m_3, m_4, m_5)$
 - Euclidean distance: $D(\mathbf{A}, \mathbf{M}) = \sqrt{\sum_{i=1}^5 (a_i - m_i)^2}$
- Main findings
 - Disclosure and transparency help narrow divergence.
 - Difference consistent with each stakeholder group's incentives and roles in company.
 - Culture and governance may be alternative mechanism.

Determinants of Divergence on Culture

Variable	Analyst– Executive	Analyst– Employee	Analyst– Executive	Analyst– Employee	Analyst– Executive	Analyst– Employee
	Type divergence	Type divergence	Cause divergence	Cause divergence	Effect divergence	Effect divergence
	(1)	(2)	(3)	(4)	(5)	(6)
Firm size	-.032*** (.003)	-.030*** (.003)	-.003 (.004)	-.016*** (.004)	-.015*** (.003)	-.011*** (.004)
ln(Firm age + 1)	.013* (.007)	.014** (.006)	-.003 (.007)	.005 (.008)	-.003 (.006)	.017** (.007)
Sales growth	.026* (.016)	.000 (.014)	-.039** (.019)	-.015 (.021)	-.002 (.015)	.011 (.017)
ROA	-.070 (.057)	-.072 (.047)	.009 (.072)	.116* (.070)	-.049 (.055)	.038 (.061)
Leverage	.032 (.022)	.042** (.020)	-.029 (.025)	-.014 (.024)	-.019 (.019)	.012 (.023)
Tangibility	.023 (.024)	.030 (.021)	-.038 (.025)	.028 (.029)	.027 (.021)	.020 (.026)
ROA volatility	.247*** (.075)	.109 (.068)	.222** (.087)	.123 (.100)	.234*** (.072)	.122 (.084)
Large institutional ownership	.027* (.014)	.006 (.012)	.030* (.016)	.026 (.017)	.032** (.013)	.049*** (.014)
Board independence	.062* (.033)	.078** (.031)	.130*** (.039)	.066 (.041)	.073** (.033)	.002 (.037)
Loss year	-.011 (.012)	-.027*** (.010)	.005 (.014)	-.014 (.015)	-.003 (.011)	-.020 (.012)
CEO duality	.001 (.000)	-.001 (.000)	-.000 (.000)	-.005 (.000)	-.001 (.000)	-.006 (.008)
CEO tenure						.001

Analysts' Perspectives and their Research Output

Variable	Recommendation (1)	Recommendation (2)	Target price (3)	Target price (4)
Tone	.127*** (.011)	.102*** (.014)	.024*** (.003)	.019*** (.003)
Tone × Number of cause and effect		.017*** (.006)		.004** (.001)
Number of cause and effect		-.012** (.005)		-.003** (.001)
Nonculture tone	.562*** (.029)	.560*** (.029)	.108*** (.008)	.108*** (.008)
ln(Report length)	.000 (.013)	.001 (.013)	-.004 (.003)	-.004 (.003)
Star analyst	-.014 (.037)	-.014 (.037)	-.004 (.008)	-.004 (.008)
Female	.010 (.044)	.011 (.044)	-.011 (.008)	-.011 (.008)
Forecast horizon	-.099** (.039)	-.099** (.039)	.002 (.008)	.002 (.008)

Information Content of Analysts' Perspectives on Culture

	CAR[-1,+1]	CAR[-1,+1]	CAR[-1,+1]	CAR[-1,+1]	CAR[-1,+1]
Variable	(1)	(2)	(3)	(4)	(5)
Tone	.258*** (.070)	.103 (.083)	.048 (.100)	.177* (.105)	.044 (.112)
Tone × High number of cause and effect		.449*** (.146)			
High number of cause and effect		-.337** (.138)			
Tone × Analyst–executive divergence			.521* (.275)		
Analyst–executive divergence			-.427* (.255)		
Tone × Analyst–employee divergence				.733** (.285)	
Analyst–employee divergence				-.185 (.264)	
Tone × Analyst–other divergence					.955** (.426)
Analyst–other divergence					-.572 (.387)

Discussion

- Generative AI as reasoning agent to construct structured knowledge
- Entity recognition $\Rightarrow$ Relation recognition $\Rightarrow$ Causal chain extraction $\Rightarrow$ Knowledge graph construction
 - Financial misstatement potential patterns
 - Representation of narrative structure
 - Managerial ability KG
 - Supply chain KG
 - Innovation KG

中国上市公司研报的企业文化分析

- 研报爬取自东方财富网站，搜集到 2025 年 9 月至 2026 年 4 月共 16,242 份研报。
- 将 PDF 研报转换为 XML 格式，并剔除表格、附录、脚注等非正文内容；随后对文本进行句子切分，并利用 C99 将句子重新聚合为语义一致的文本片段。
- 直接利用 GPT-5.4 识别与企业文化相关的文本片段，包括文化描述、价值观表达以及管理风格相关叙事。
- 模仿本文提示词框架，设计中文提示词。对文化片段进行企业文化类型识别、文化成因与结果抽取以及因果三元关系构建。标准化分类沿用文中六大文化类型（协作与以人为本、客户导向、创新与适应性、诚信与风险管理、绩效导向、其他），以及标准化的 18 个文化成因和 17 个文化结果。

示例

在业务演进路径上，公司正由传统光模块厂商向“高端光互联解决方案提供商”升级：一方面，公司持续推进产品向更高速率（800G→1.6T）迭代，提升产品单价与技术壁垒；另一方面，公司通过提前锁定上游资源与扩充产能（如预付款与在建工程显著提升），强化交付能力，以适应 AI 算力需求快速增长带来的供给压力。整体来看，公司增长逻辑已由“通信流量驱动”向“AI 算力驱动”切换，行业属性由周期性通信设备向高景气算力基础设施演进。

从产业链定位来看，公司处于 AI 算力产业链中游的关键互联环节，上游连接光芯片、光器件及材料供应商，下游服务于云厂商、互联网公司 & AI 基础设施建设方。随着 AI 集群规模持续扩大及算力密度提升，网络互联的重要性不断增强，光模块已由“配套组件”升级为“性能瓶颈核心环节”。在此背景下，公司有望受益于行业景气提升及自身技术与客户优势，持续实现规模扩张与盈利能力提升，逐步形成“高景气驱动+高端产品结构升级”的成长路径。

- AI 算力需求扩张（颠覆性技术变革）⇒ 持续技术迭代与研发投入（创新与适应文化）
- 受益于自身技术与客户优势（创新与适应文化、客户导向文化）⇒ 实现规模扩张与盈利能力提升（市场份额与成长提升）

描述性统计

中国上市公司分析师研报中的企业文化识别统计

分析师研报	研报数量	公司-年份	公司数量
原始研报	16,242	4,120	2,876
识别出文化	4,286	2,158	1,642
识别出文化成因	1,835	1,267	1,085
识别出文化结果	2,974	1,684	1,326
识别出成因或结果	3,158	1,742	1,381
同时识别出成因与结果	1,651	1,183	1,024

企业文化因果关系

